MEDIUM ALGEBRA

Algebra

07

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the xy-plane. What is the slope of line j?

A. $\frac{7}{17}$

B. $\frac{17}{7}$

C. 4

D. 17

Q2 ALGEBRA

A candle is made of 17 ounces of wax. When the candle is burning, the amount of wax in the candle decreases by 1 ounce every 4 hours. If 6 ounces of wax remain in this candle, for how many hours has it been burning?

A. 3

B. 6

C. 24

D. 44

Q3 ALGEBRA

$$gm = \text{-}0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

A. 0.05

B. 12.1

C. 20

D. 242.0

Q4 ALGEBRA

A bus traveled on the highway and on local roads to complete a trip of 160 miles. The trip took 4 hours. The bus traveled at an average speed of 55 miles per hour mph on the highway and an average speed of 25 mph on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

A. $55x + 25y = 4$
$x + y = 160$

B. $55x + 25y = 160$
$x + y = 4$

C. $25x + 55y = 4$
$x + y = 160$

D. $25x + 55y = 160$
$x + y = 4$

Q5

GRID-IN ALGEBRA

A number x is at most 17 less than 5 times the value of y. If the value of y is 3, what is the greatest possible value of x?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ALGEBRA

When line n is graphed in the xy-plane, it has an x-intercept of $-4, 0$ and a y-intercept of $0, \frac{86}{3}$. What is the slope of line n?

A. $\frac{3}{344}$

B. $\frac{6}{43}$

C. $\frac{43}{6}$

D. $\frac{344}{3}$

Q7 ALGEBRA

$$a(3-x)-b=-1-2x$$

In the equation above, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b ?

A. $a=2$ and $b=1$

B. $a=2$ and $b=7$

C. $a=-2$ and $b=5$

D. $a=-2$ and $b=-5$

Q8 ALGEBRA

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

A. -4

B. 8

C. 42

D. 44

Q9 ALGEBRA

$$4x + 5y = 100$$
$$5x + 4y = 62$$

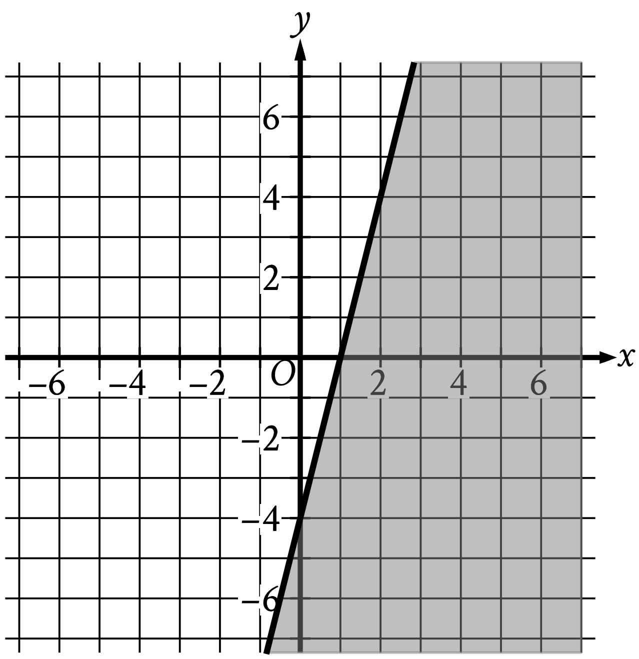
If the system of equations above has solution (x, y), what is the value of $x + y$?

A. 0

B. 9

C. 18

D. 38



- The boundary of the inequality is a solid line.
    - The line slants sharply up from left to right.
    - The line passes through the following points:
        - (0 comma negative 4)
        - (1 comma 0)
        - (2 comma 4)
- The area below and to the right of the boundary is shaded.

The shaded region shown represents the solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

A. -5, -6

B. -2, 5

C. 1, 4

D. 6, - 2